Project Smart Trigger Design Documentation for ECE/CS 3992

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***Abstract*—This document describes the design of the Smart Trigger, a modular RFID-based access-control system for the AR-15 platform. A passive 125kHz RFID tag worn by the authorized user is read by an antenna embedded in a custom 3D-printed grip. A Raspberry Pi Pico microcontroller validates the tag against an authorized list and drives a micro stepper motor that moves a blocking member out of the fire control group. The document covers the system architecture, the hardware components and their integration onto a custom PCB, the CAD modeling of the grip and actuator mounts, and the firmware processes that tie the system together. It closes with a look at how the design could grow through an off-device logging server and web application, and what that complexity would cost in power, security, and development effort.**

I. Introduction

II. System Architecture

A. Overview

The Smart Trigger is a self-contained, battery-powered embedded system that adds an electronic authorization step to the mechanical operation of an AR-15. Functionally, it is a closed loop with three stages: Sense, Decide, Act.

B. Subsystems

C. Design Requirements

D. Key Architectural Decisions

III. Hardware Components and Integration

A. Mechanical Interaction

The mechanical design has three parts:

1) a **custom grip** that replaces the factory AR-15 grip and houses the electronics, battery, and RFID antenna
2) an **actuator mount** that locates the micro stepper in the fire control pocket of the lower receiver
3) a **blocking member** driven by the motor that physically prevents the fire control group from releasing the hammer

Part Number	Short Name	Short Description
Pi Pico W	Pico	Microcontroller and central brain
SAA1042	Motor Driver	Stepper motor driver
RDM6300	RFID Reader	RFID reader and translator
ANT1	Antenna	Custom wound antenna tuned to 125kHz
M1	Motor	Stepper motor to rotate shaft
R1	R	47kΩ resistor
C1	C	100 nF capacitor

To understand the innovation behind the blocking member takes some understanding of the original function of the AR-15 fire control group(trigger, hammer, sear, and safety). To fire a round the trigger must be pulled, the angle the trigger is brought to in its movement releases the hold it had on the spring loaded hammer. The freed hammer slams against onto a hook called the disconnector which holds the hammer until the trigger is back to its ready position it was in before firing the last round. The safety on the AR-15 is a physical barrier to the trigger. When is the safe position the trigger is physically blocked from movement thus preventing the hammer from moving and the overall firing of the gun. The design of our safety takes advantage of the mechanical blocker of the AR-15's native safety selector. To create the most widely applicable design we focused on locking the native safety in place until an authorized user is recognized.

B. Component Selection
C. Main Board (PCB)
D. RFID Board (RDM6300)
E. Micro Stepper Motor Driver Board (SAA1042)
F. Integration and Test Plan

IV. CAD Modeling and Design

A. Scope of the Mechanical Design
B. Custom Grip
C. Actuator Mount and Blocking Member
D. Materials and Print Parameters
E. Design Iteration

V. Software Processes

A. Development Environment
B. Firmware Architecture
C. Control State Machine
D. RFID Processing
E. Actuator Control
F. Software Testing
G. Expansion: Logging Server and Web Application

VI. Conclusion

REFERENCES